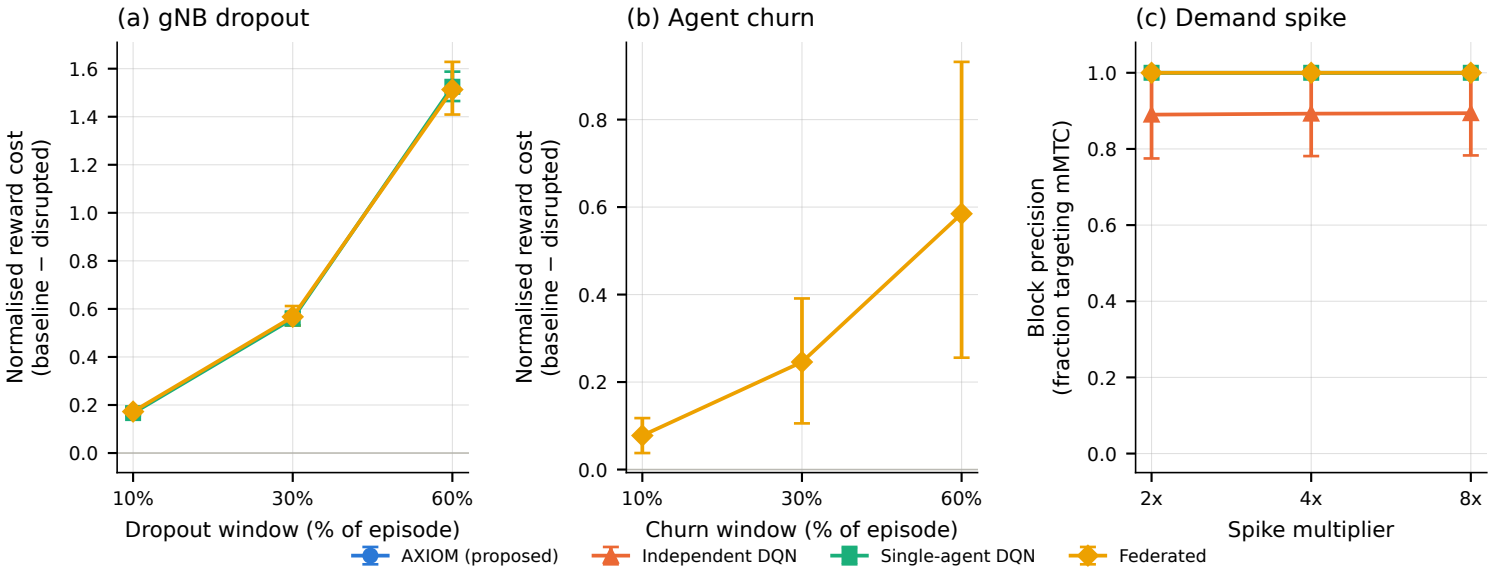
M4: disruption cost vs. severity, 10 seeds/arm
(a) gNB dropout
(b) Agent churn
(c) Demand spike
Normalised reward cost
(baseline − disrupted)
Block precision
(fraction targeting mMTC)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
10%
30%
60%
2x
4x
8x
Dropout window (% of episode)
Churn window (% of episode)
Spike multiplier
AXIOM (proposed)
Independent DQN
Single-agent DQN
Federated